# To‘g‘ridan-to‘g‘ri va statik marshrutlar

## Direct Routes and Static Routes

Asosiy atamalar

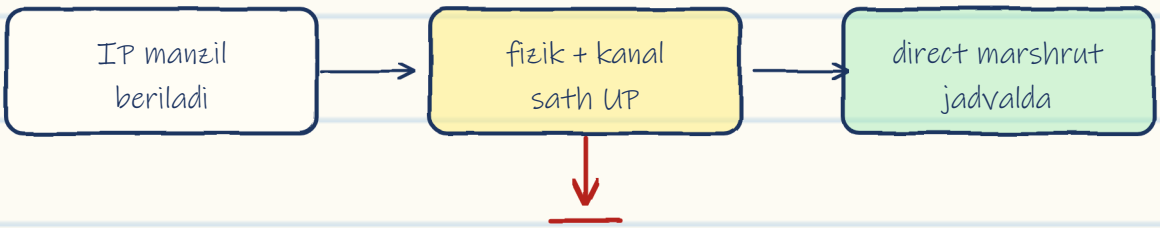
- To‘g‘ridan-to‘g‘ri marshrut — interfeys yoqilganda o‘zi paydo bo‘ladi (direct)
- Statik marshrut — qo‘lda yoziladi, o‘zgarmaydi
- Next hop — paket keyingi qaysi qurilmaga berilishi
- Sukut marshruti — 0.0.0.0/0 — mos yo‘l topilmasa ishlatiladi
- Preference — marshrut manbasiga ishonch darajasi, statikda 60
- Null 0 — paketni yo‘q qiluvchi «qora tuynuk» interfeysi

## 1. Direct marshrut

Interfeysga IP berilsa — marshrut o‘zi paydo bo‘ladi, qo‘lda yozilmaydi.

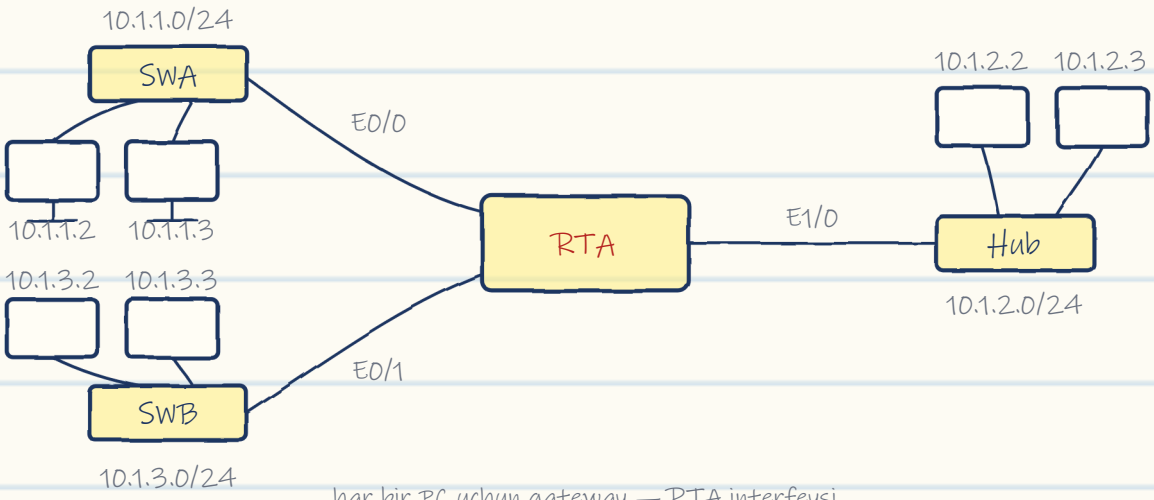
- shart — fizik va kanal sathi ikkisi ham UP
- DOWN — marshrut jadvalga qo‘yilmaydi

qo‘lda sozlash kerak emas




1-rasm. Direct marshrut interfeys holatiga bog‘liq

## 2. LAN'lar orasida marshrutlash

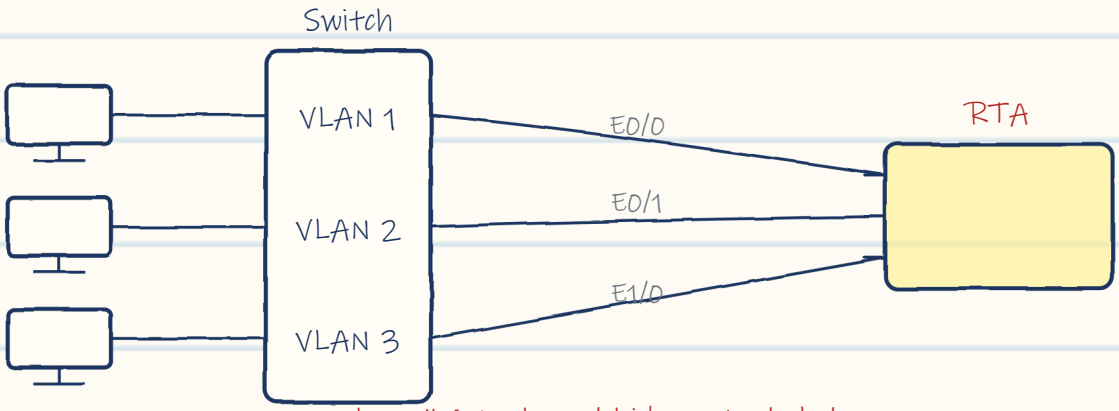



har bir PC uchun gateway — RTA interfeysi

2-rasm. Uch interfeys — uch segment, hammasi direct marshrut

## 3. VLAN'lar orasi: qimmat yechim

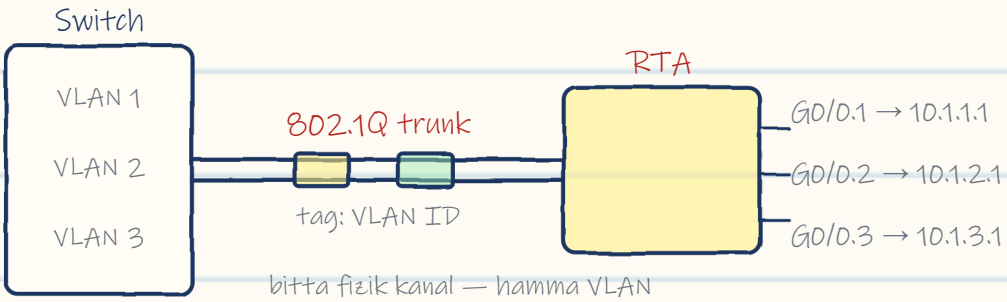
Har VLAN — o'z IP segmenti, ya'ni routerda alohida fizik port.




har VLAN uchun alohida port + kabel

3-rasm. VLAN ko'p bo'lsa — port va kabel yetmaydi

## 4. Bir qo‘li marshrutlash (802.1Q + subinterfeys)

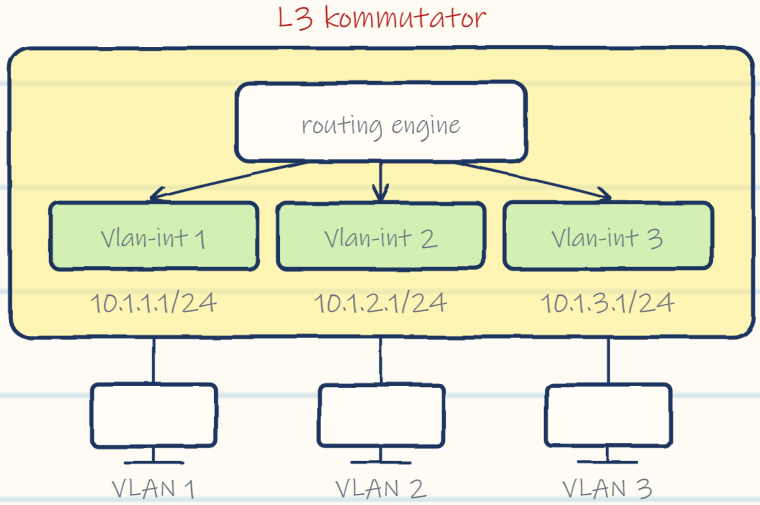



```
[RTA]interface GigabitEthernet0/0.2
[RTA-GE0/0.2]vlan-type dot1q vid 2
[RTA-GE0/0.2]ip address 10.1.2.1 255.255.255.0
```

4-rasm. Single-arm: bitta port, uch subinterfeys

Qoida: trunk hamma VLAN trafigini olib yuradi — keng polosali kanal tanlanadi. G0/0.1 default VLAN 1, tag qo‘yilmaydi.

## 5. L3 kommutatorda VLAN’lar orasi




5-rasm. Har VLAN uchun VLAN-interfeys, marshrut apparatda

- VLAN-interfeys — router porti kabi, IP oladi
- apparat — yuqori tezlik, katta o'tkazuvchanlik
- trunk kanalidagi kechikish va nobarqarorlik yo'q

## 6. Statik marshrut buyrug'i

```
ip route-static 10.4.0.0 24 10.3.0.2
```

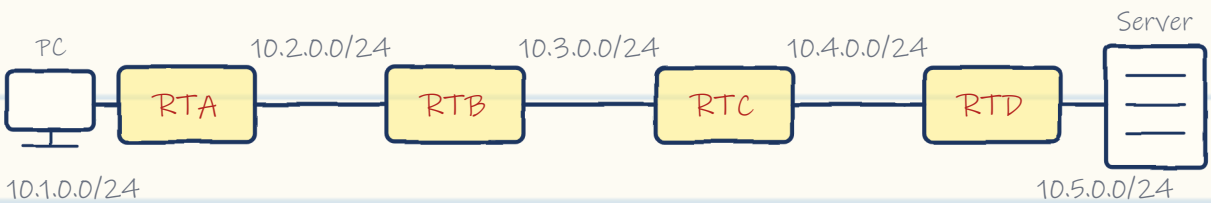
| | |
|---|---|
| dest + mask | qayerga borish kerak |
| next-hop | qo'shni IP — ulangan bo'lsin |
| interfeys | P2P yoki null 0 uchun |
| preference | 1–255, standart 60 |

6-rasm. Broadcast interfeysda next-hop ham ko'rsatiladi

Kamchilik: statik marshrut topologiya o'zgarishiga moslashmaydi — kanal tushsa, administrator qo'lda tuzatadi. Shuning uchun faqat oddiy tarmoqlarda ishlatiladi.

## 7. Misol: uchdan-uchga statik marshrut

Paket hop-by-hop uzatiladi — har routerda hamma segment yozilishi kerak.



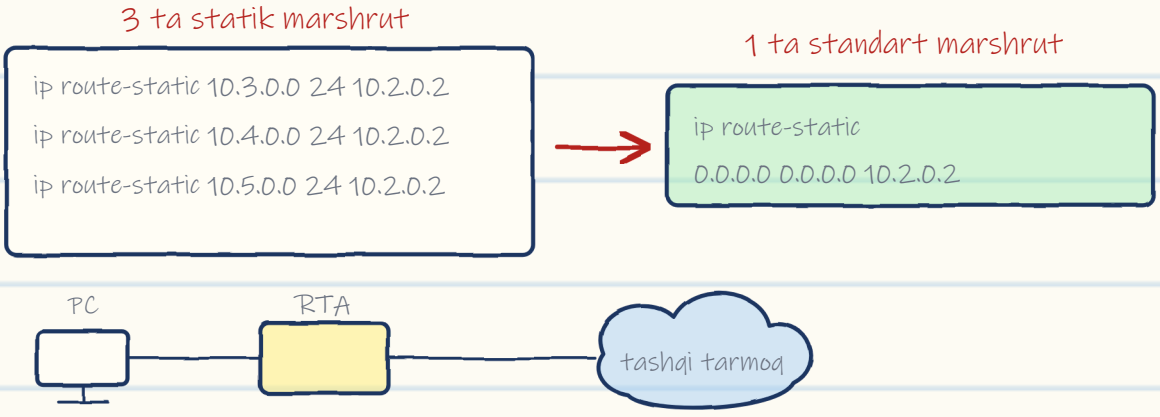

[RTB] ip route-static 10.4.0.0 255.255.255.0 10.3.0.2

har bir routerda hamma segmentga marshrut

7-rasm. PC va Server gaplashishi uchun 4 routerda marshrut

Qoida: next-hop bevosita ulangan va yetib boradigan bo'lsin — aks holda router kanal sathi manzilini olmaydi.

## 8. Standart marshrut

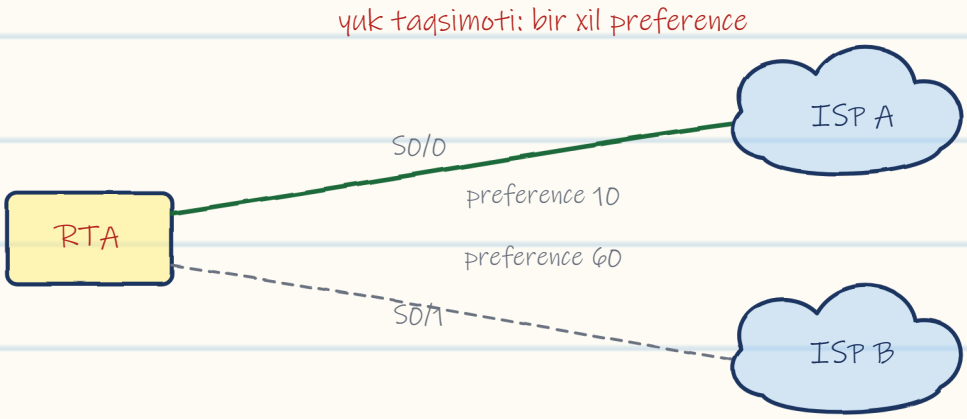



stub tarmoq — bitta chiqish

8-rasm. Jadval kichrayadi, tanlash tezlashadi

- standart marshrutni OSPF · IS-IS · RIP ham hosil qila oladi
- Internetdagi routerlarning deyarli hammasida default bor

## 9. Yuk taqsimoti va zaxira

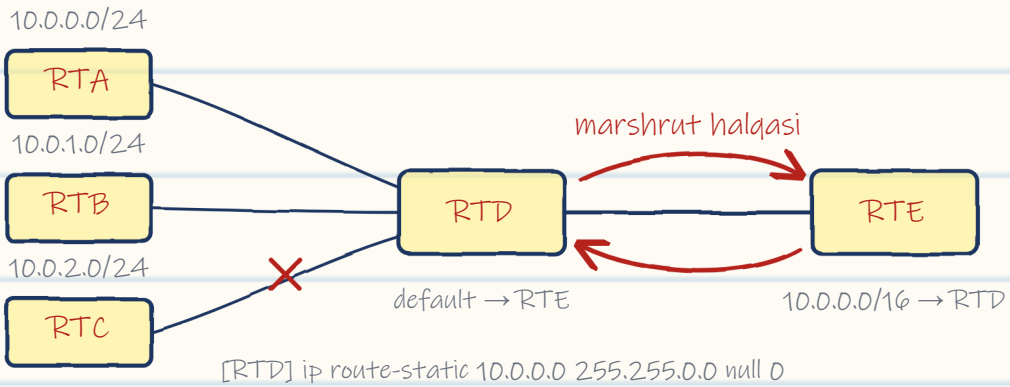



zaxira: kichik preference = asosiy yo'l

9-rasm. S0/0 tushsa — trafik S0/1 orqali ketadi

| | | |
|---|---|---|
| preference | bir xil → yuk taqsimoti | turlicha → zaxira |
| standart | statik marshrut = 60 | diapazon 1–255 |

## 10. Qora tuynuk marshrut (null 0)




[RTD] ip route-static 10.0.0.0 255.255.0.0 null 0

paket null 0 ga tashlanadi — halqa yo'q

10-rasm. RTC kanali uzilsa, RTD paketni null 0 ga beradi

> Eslatma: null 0 interfeysiga IP berib bo'lmaydi. Yolg'iz o'zi foydasiz — u umumlashtirilgan marshrut bilan birga ishlaydi.

## Xulosa

| tur | qanday | izoh |
|---|---|---|
| direct | IP + UP/UP | avtomatik |
| static | next-hop yoki p2p | pref 60 |
| default | 0.0.0.0 0.0.0.0 | stub tarmoq |
| null 0 | qora tuynuk | halqaga qarshi |

11-rasm. To'rt xil marshrut — bitta jadvalda

- Direct — IP beriladi, ikki sath UP bo'lsa jadvalga tushadi
- VLAN'lar orasi: 3 port · 1 trunk + subinterfeys · L3 kommutator
- Statik marshrut qo'lda, topologiya o'zgarsa o'zi tuzatilmaydi
- Next-hop bevosita ulangan bo'lishi shart
- 0.0.0.0/0 — standart marshrut, stub tarmoqda
- null 0 — qora tuynuk, marshrut halqasini to'xtatadi

Tekshirish

1. Direct marshrut sharti? — fizik va kanal sathi UP
2. Single-arm nechta fizik port? — 1, trunk
3. Statik marshrut standart preference? — 60 (1–255)
4. Standart marshrut yozilishi? — 0.0.0.0 0.0.0.0
5. null 0 nima uchun? — halqaga qarshi, paket tashlanadi